STOCHASTIC MODELING

MODULE 4 | LESSON 1

INTEREST RATES MODELS

Reading Time	40 minutes
Prior Knowledge	Vasicek model, Monte Carlo
Keywords	Vasicek, CIR, Simulation

In this lesson we will look at interest rate models, a first step towards augmenting our stochastic volatility jump-difussion model. First, we will revisit the Vasicek (1977) model. Then, we will study Cox-Ingersoll-Ross model of 1985, famously known as CIR. Finally, we will use Monte Carlo to compare the different short rate paths simulated by both models.

As usual, let's start with the import the necessary libraries:

In [1]:
```
import matplotlib.pyplot as plt
import numpy as np
```

1. Vasicek (1977) Model

We already know the form of Vasicek (1977) SDE and how to implement it in Python from previous course. Please go back to those if you need a refresher on this.

The risk-neutral process for short rates in Vasicek (1977) is given by:

$$ \ \begin{equation*} dr_t = k(\theta-r_t)dt + \sigma dz_t \end{equation*} $$
where $dz_t = \sqrt{dt} z$, with $z \sim \mathcal{N}(0,1)$.

In [2]:
```
def vasicek(r0, k, theta, sigma, T, N, M):
    dt = T / N
    rates = np.zeros((N, M))
    rates[0, :] = r0
    for j in range(M):
        for i in range(1, N):
            dr = (
                k * (theta - rates[i - 1, j]) * dt
                + sigma * np.sqrt(dt) * np.random.normal()
            )
            rates[i, j] = rates[i - 1, j] + dr
    return rates
```

Using this function, and some exogenously given parameters for k, θ, σ and r_0, let's simulate 100 price paths that take 100 steps in 1 year:

- $r_0 = 2.3\%$
- $k = 0.2$
- $\theta = 0.01$
- $\sigma = 0.012$

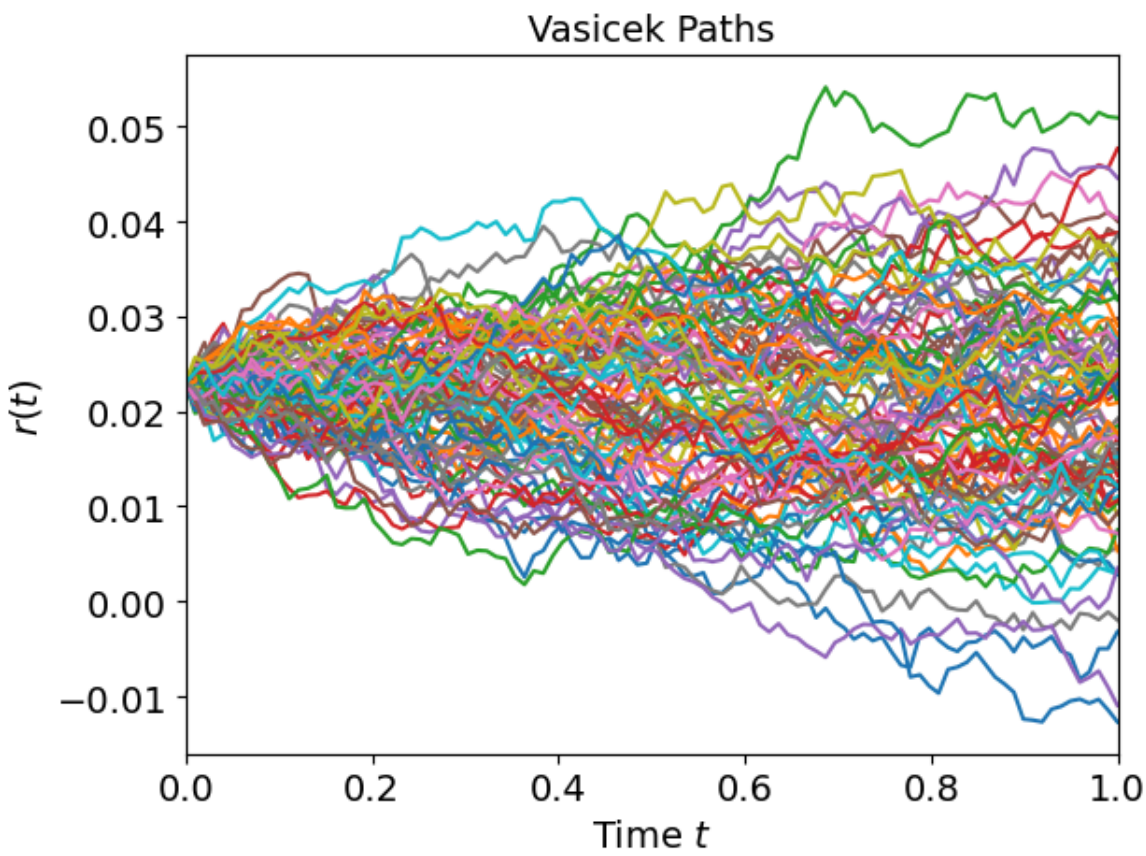
In [3]:
```
M = 100  # Number of paths for MC
N = 100  # Number of steps
T = 1.0  # Maturity
r0 = 0.023
k = 0.20
theta = 0.01
sigma = 0.012
t = np.linspace(0, T, N)

rates = vasicek(r0, k, theta, sigma, T, N, M)
```

And plot the different paths...

In [4]:
```
for j in range(M):
    plt.plot(t, rates[:, j])

plt.xlabel("Time $t$", fontsize=14)
plt.ylabel("$r(t)$", fontsize=14)
plt.title("Vasicek Paths", fontsize=14)
axes = plt.gca()
axes.set_xlim([0, T])
plt.xticks(fontsize=14)
plt.yticks(fontsize=14)
plt.tight_layout()
plt.show()
```



2. Cox-Ingersoll-Ross (1985) Model

Now, let's undergo the same process with the CIR (1985) model, which is not so familiar to us, although it simply involves a small switch from Vasicek's SDE.

CIR (1985) model SDE has the following form:

$$ \ \begin{equation*} dr_t = k_r(\theta_r-r_t)dt + \sigma_r \sqrt{r_t} dz_t \end{equation*} $$

So, with just a couple small tweaks to the Vasicek (1977) function, we can get the one for CIR (1985):

In [5]:
```
def cir(r0, k, theta, sigma, T, N, M):
    dt = T / N
    rates = np.zeros((N, M))
    rates[0, :] = r0
    for j in range(M):
        for i in range(1, N):
            dr = (
                k * (theta - rates[i - 1, j]) * dt
                + sigma
                * np.sqrt(dt)
                * np.sqrt(np.maximum(rates[i - 1, j], 0))
                * np.random.normal()
            )
```

```
            rates[i, j] = rates[i - 1, j] + dr
    return rates
```

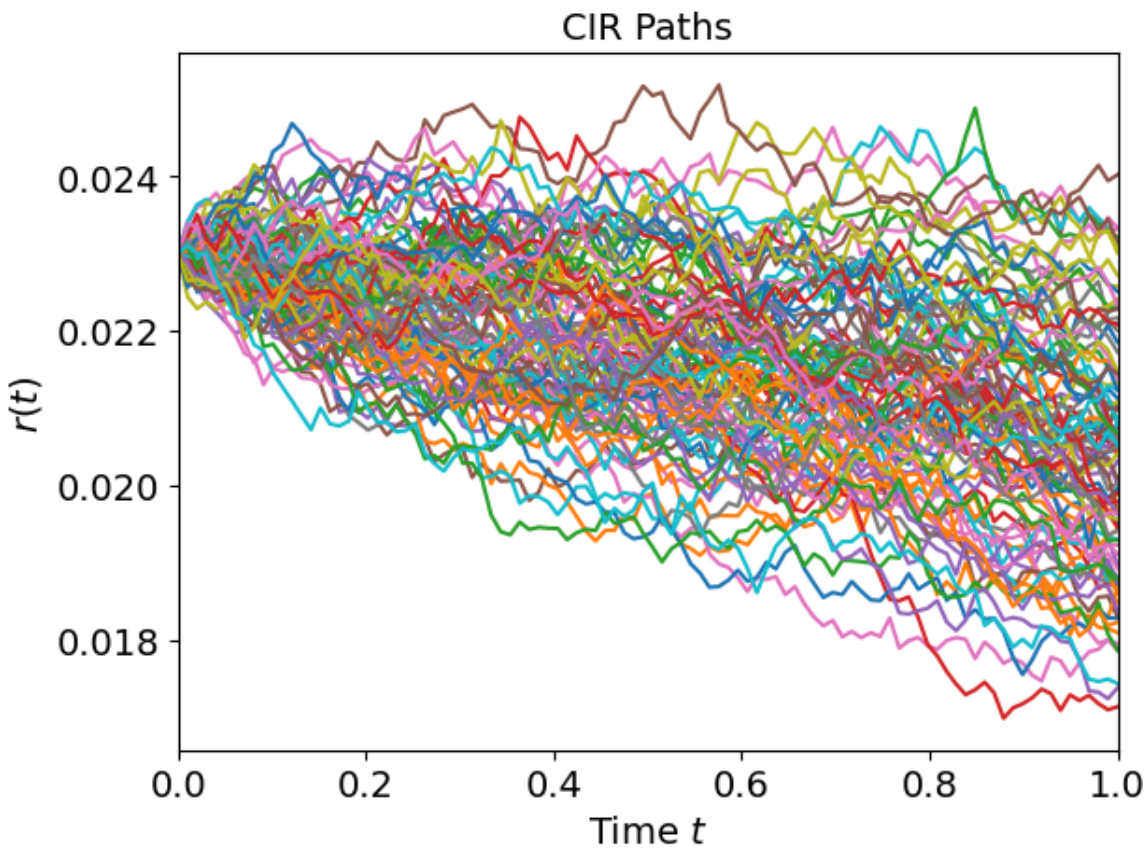
Note that we have included a truncation on the $\sqrt{r_t}$ component just to ensure that we do not get an error for $r_t<0$.

The simulated rates under CIR look like the following:

In [6]:
```
rates_cir = cir(r0, k, theta, sigma, T, N, M)
```

In [7]:
```
for j in range(M):
    plt.plot(t, rates_cir[:, j])

plt.xlabel("Time $t$", fontsize=14)
plt.ylabel("$r(t)$", fontsize=14)
plt.title("CIR Paths", fontsize=14)
axes = plt.gca()
axes.set_xlim([0, T])
plt.xticks(fontsize=14)
plt.yticks(fontsize=14)
plt.tight_layout()
plt.show()
```



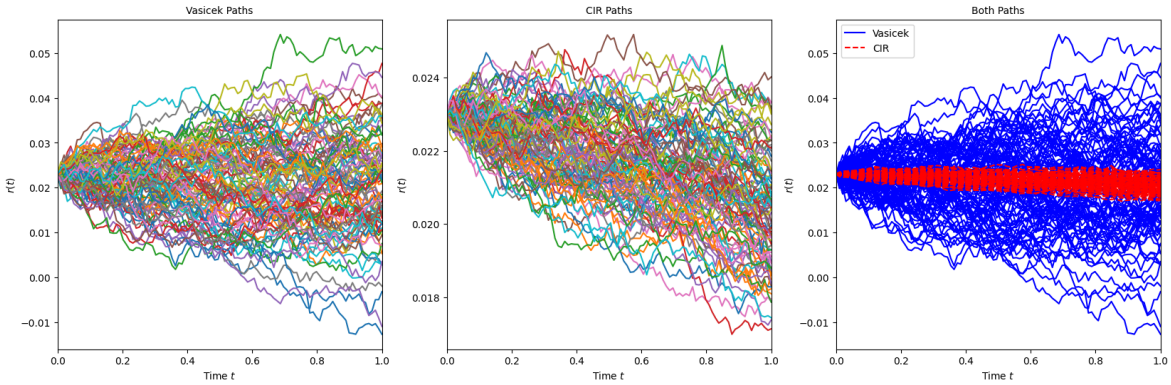
3. Comparison of Vasicek (1977) and CIR (1985)

Finally, let's compare the short rate paths generated by both models. In order to do so, let's graph them against each other:

In [8]:
```
plt.figure(figsize=(20, 6))
plt.subplot(131)
plt.plot(t, rates[:, :])
plt.xlabel("Time $t$", fontsize=10)
plt.ylabel("$r(t)$", fontsize=10)
plt.title("Vasicek Paths", fontsize=10)
axes = plt.gca()
axes.set_xlim([0, T])
plt.xticks(fontsize=10)
plt.yticks(fontsize=10)

plt.subplot(132)
plt.plot(t, rates_cir[:, :])
plt.xlabel("Time $t$", fontsize=10)
plt.ylabel("$r(t)$", fontsize=10)
plt.title("CIR Paths", fontsize=10)
axes = plt.gca()
axes.set_xlim([0, T])
plt.xticks(fontsize=10)
plt.yticks(fontsize=10)

plt.subplot(133)
plt.plot(t, rates[:, 1:], "b")
plt.plot(t, rates[:, 0], "b", label="Vasicek")
plt.plot(t, rates_cir[:, 1:], "r--")
plt.plot(t, rates_cir[:, 0], "r--", label="CIR")
plt.legend()
plt.xlabel("Time $t$", fontsize=10)
plt.ylabel("$r(t)$", fontsize=10)
plt.title("Both Paths", fontsize=10)
axes = plt.gca()
axes.set_xlim([0, T])
plt.xticks(fontsize=10)
plt.yticks(fontsize=10)
plt.show()
```



Is this a fair comparison of both models? No it's not!

In [9]:
```
from IPython.display import VimeoVideo

# Bigger video
VimeoVideo("1121446507", h="3298dbabb7", width=700, height=450)
```

Out[9]:

Does this make sense? Well, if you look at the actual SDEs and think about it, it does! Basically because, the way we are currently doing things, the volatility term of $\sigma \sqrt{r_t}$ in CIR is, by construction, always smaller than the volatility term in Vasicek (simply σ). Essentially, the volatility term in CIR is the volatility term in Vasicek times a number lower than 1, which results in a strictly lower volatility term.

Is this correct then? No, it is not! For starters, there is a reason why we named the volatility terms differently (σ and σ_r): these are not the same parameter. Secondly, we need to carefully consider how we treat $\sqrt{r_t}$ since this is a squared-root process and discretized versions have to be considered.

How do I get these parameters then? That is precisely what we will do in the next lesson, perform market calibration of interest models. As you will see there, we will get completely different estimates for σ and σ_r, as we should. \

Code Task

Implement `extract_terminal_rates(model_func, r0, k, theta, sigma, T, N, M, seed=None)`

Goal
Write a helper that runs a provided short-rate simulator (`cir` or `vasicek`) and returns a 1D NumPy array of terminal rates (last time step) with shape `(M,)` .

Requirements

- If seed is not None, set reproducibility by calling `np.random.seed(seed)` before calling `model_func`.
- Call `model_func(r0, k, theta, sigma, T, N, M)` to obtain a rates array of shape `(N, M)`.
- Return a 1D NumPy array of length `M` containing the last row (terminal rates).
- Do not modify global state besides the optional `np.random.seed(seed)` call. Do not change `cir` or `vasicek`.

Function signature

```
def extract_terminal_rates(model_func, r0, k, theta, sigma, T, N,
M, seed=None):
    ...
```

In [10]:
```
### Starter code
import numpy as np

def extract_terminal_rates(model_func, r0, k, theta, sigma, T, N, M, seed
    """
    Run the provided short-rate simulator (cir or vasicek) and return a 1
    of terminal rates with shape (M,). If `seed` is given, set np.random.
    before calling the simulator for reproducibility.
    """
    # TODO:
    # 1) If seed is not None: np.random.seed(seed)
    # 2) Call model_func(r0, k, theta, sigma, T, N, M) to get an (N, M) a
    # 3) Return rates[-1, :] as a 1D NumPy array of length M
    raise NotImplementedError("Implement extract_terminal_rates as specif
```

4.Conclusion

In this lesson, we briefly revisited the Vasicek (1977) interest rate model and enhanced it by introducing the Cox-Ingersoll-Ross (1985) model. For these two models, we have applied Monte-Carlo methods to simulate the different short rate paths that each generates, only to realize that without proper parameter calibration, a comparison of the models does not make much sense from a practical perspective. In the next lesson, we will go over the calibration process for interest rate models.

References

- Cox, John C., et al. "An Intertemporal General Equilibrium Model of Asset Prices." *Econometrica: Journal of the Econometric Society*, 1985, pp. 363-384.

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